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Problem Description

Overview: As a result of developing an Intelligent Autonomous Agent for optimising multi-stop logistics and a constrained 2D warehouse grid, we will be tasked with solving a Constrained Multi-Goal routing problem that involves balancing item priorities, spatial efficiencies and strict energy constraints, rather than just basic pathfinding.

The agent's mission is defined by five core behaviors:

1. Grid Navigation: Traversal of a discrete coordinate system.
2. Strategic Collection: Harvesting multiple items scattered across non-adjacent cells
3. Prioritized Delivery: Transporting the payload to fixed "Depot" or "Delivery Zone".
4. Obstacle Navigation: Real-time avoidance of static structural barriers
5. Energy Management: Executing the entire mission before the battery level reaches zero.

Complexity Level:

To better define the "L3" complexity we can use known challenges in computer science. As to why the complexity is L3:

- → Multiple items → combinatorial explosion (2^k subsets)
- → Energy feasibity constraint.
- → weighted objective function
- → Must plan order of pickups strategically.

This is computationally similar to a constrained Travel Salesperson Problem (NP hard problem)

Assumptions & Limitations:

- → Static obstacles (no moving agents)
- → Deterministic motion.
- → Single Robot.
- → Known full map.
- → Energy consumption per move = 1 unit.

Future L4 extension:

- → Dynamic obstacles → replanning.
- → Multi-agent coordination
- → Stochastic energy usage.

Environment and PEAS

Environment Description

Property	Type	Justification
1. Observable	Fully Observable	Entire grid map known
2. Deterministic	Yes	Actions have predictable outcomes
3. Episodic	No (sequential)	Current move affects future states
4. Static	Yes	Map does not change
5. Discrete	Yes	Finite grid cells
6. Single Agent	Yes	Only one robot

PEAS

Performance Measure ⇒
- maximize : Total priority collected
- minimize : Total path cost
- Subject to : Energy ≥ 0

We define objective : $U = \sum_{j \in collected} w_j - \lambda . \text{path length}$

where w_j = priority weight and $\lambda > 0$ balances travel cost

Environment :-

1. Grid (G) : $G \subseteq \mathbb{Z}^2$, size $N \times N$, where $10 \leq N \leq 50$
2. Obstacle set : $O \subset G$
3. Item set : $I = \{i_1, i_2, i_3 \ldots i_k\}$
4. Delivery zone : $d \in G$

Actuators :-

1. move up
2. Move down
3. Move left
4. move right
5. Pick
6. Deliver

Sensors :-

1. Current position (x, y)
2. Remaining energy
3. Item presence at cell
4. Full grid map.

SEARCH FORMULATION : (L3 COMPLEXITY)

- The warehouse robot problem is modeled as a search and optimisation problem through space where a robot must navigate a grid, collect items and deliver them while avoiding obstacles.

State space (S) :

It is the formal, mathematical set of all possible states (or configurations) a system can occupy along with the transitions between them

A state $S : \{(x, y), C, E\}$

- (x, y) are the current coordinates of the robot on the grid scaling from 10×10 to 50×50
- C is the set representing which items are collected eg. $C = \{i_2, i_3\}$ means the robot has collected items i_2 and i_3. C is the subset of I which is the set of all items
- E is the energy of the robot at that state. Let E_{max} be maximum energy. Then $0 \leq E \leq E_{max}$

Initial state (S_0) :

Initial state $S_0 = \{(x_0, y_0), \phi, E_{max}\}$

Where (x_0, y_0) are the initial coordinates of the robot. The set C is denoted by ϕ as the robot has not yet collected any items

Actions / Preconditions and Effects :- (A)

These are the set of all active movements a robot can make.

It includes A - Move { Up, Down, Right, Left }

The preconditions of an action are :-

1) Robot can only make movements that allow it to stay inside a grid boundary

2) Robot has to avoid all static obstacles in its path, meaning it cannot cross over them

3) Energy E must be greater than or equal to 1. $E \geq 1$

Transition Model (T)

If A = Move { Up } , new state $S = \{(x, y+1), C', E - cost\}$

If A = Move { Down }, new state $S = \{(x, y-1), C', E - cost\}$

If A = Move { Left } , new state $S = \{(x-1, y), C', E - cost\}$

If A = Move { Right } , new state $S = \{(x+1, y), C', E - cost\}$

Item collection : If during the action A, the moving robot moves through an uncollected object i_j, collection set

$C' = C \cup \{i_j\}$

Energy of new state E' is defined as E - cost needed to take action A.

Goal Test (Goal(s)) :-

Returns true when an objective is reached.

A state is a goal state if

$(x, y) = D$, that is robot has reached delivery location

Path cost function :- ($g(n)$)

cost of reaching a specific state S.

$g(n) = \Sigma$ step cost $\times$ weight, where picking up heavier objects can increase energy cost per move

optimisation problem :-

Reward can be defined as $g(R) = W_i$ for each item i

we define 2 quantities per state

Energy used so far $gE(s) = E_{max} - E$

Reward collected so far $gR(s) = \Sigma W_i$ for each item collected

→ we must optimise the robot to maximise the reward while simultaneously minimising energy used.

State space analysis $|S|$:-

This includes the number of reachable states for the robot

Let N be no of coordinates the robot can reach.

Let O be no of obstacles on grid.

$N = (\text{Width} \times \text{Height} - O)$

Let the no of items be K. No of possible item subsets 2^K

No of energy levels $= E_{max} + 1$

So upper bound on total states

$$|S| = N \times 2^K \times (E_{max} + 1)$$

grows exponentially with no of items K and linear relative to no of grid states.

Dominance Rule (Key part of Pareto search)

In our ~~Par~~ Pareto based search, multiple labels exist for same configuration

For 2 states S1 and S2 with same configuration ((x, y) & C)

S1 dominates S2 if

~~(greater)~~) $gE(S_1) \leq gE(S_2)$ and $gR(S_1) \geq gR(S_2)$

and 1 inequality is strict.

If S1 dominates S_2, S_2 is discarded

This dominance pruning reduces no of explored states ~~significan~~ significantly.

Heuristic Design (using Pareto / dominance-pruned best first search)

⇒ For L3 complexity :-

- Given grid $R \times C$ with static obstacles
- Items :- $I = \{1, 2, \dots K\}$, location; $loc(i) = (x_i, y_i)$
- Delivery location :- $D = (x_d, y_d)$
- A search node/state

$$n \equiv s = \langle (x, y), C, e \rangle$$

where $C \subseteq I$ is the collected set, e remaining energy

We define the cost objective for admissibility proofs as

: Each move costs one energy

: we want a lower bound on the min. number of moves still required to reach delivery cell D and finish.

Let $h^*(n)$ denote the true optimal remaining cost (minimum remaining no. of moves) from n to any goal state.

- Goal predicate (used in our formulation):

$$Goal(s) \equiv ((x, y) = D)$$

Feasibility with energy is enforced by preconditions, not the heuristic

Heuristic 1: Manhattan to Delivery

Definition :- $h_1(n) = |x - x_d| + |y - y_d|$, this is the Manhattan distance from current position to delivery.

Admissibility proof :-

Let $d^*(n, D)$ be the true shortest path length from (x, y) to D in the grid with obstacles.

In a 4 connected grid, any path from (x, y) to D must change:

- row by at least $|x - x_d|$
- column by at least $|y - y_d|$

Each move changes row or column by at most 1 so any path has length at least $|x - x_d| + |y - y_d|$

obstacles can only make the path longer, never shorter

$\therefore$ $\boxed{h_1(n) \leq d^*(n, D) = h^*(n)}$ So h_1 is admissible

• Consistency Analysis

A heuristic is consistent if for every state n and successor n' reached by one action with cost 1:

$$h(n) \leq 1 + h'(n)$$

In a 4-neighbourhood grid one step changes Manhattan distance to the fixed goal by at most 1, thus:-

$$h_1(n') \geq h_1(n) - 1 \Rightarrow h_1(n) \leq 1 + h_1(n')$$

so h_1 is consistent

Heuristic 2 (Landmark Lower Bound using Grid distances)

Small set of landmark cells $L = \{l_1, l_2 \ldots l_m\}$

shortest path distances with obstacles from each landmark to every free cell using BFS: $dist(l_j, v) \; \forall v \in F$

standard in pathfinding, known as ALT heuristic (A*, Landmarks, Triangle inequality)

Definition :- For node n at position $p = (x, y)$:

$$h_2(n) = \max_{l \in L} |dist(l, D) - dist(l, p)|$$

Admissibility Proof :-

using Triangle Ineq :-

$$dist(p, D) \geq |dist(l, D) - dist(l, p)|$$

This follows from

$$dist(l, D) \leq dist(l, p) + dist(p, D) \Rightarrow dist(p, D) \geq dist(l, D) - dist(l, p)$$

and similarly

$$dist(l, p) \leq dist(l, D) + dist(p, D) \Rightarrow dist(p, D) \geq dist(l, p) - dist(l, D)$$

combine :-

$$dist(p, D) \geq |dist(l, D) - dist(l, p)|$$

Taking max over landmarks preserves the lower bound

$$dist(p, D) \geq \max_{l \in L} |dist(l, D) - dist(l, p)| = h_2(n)$$

But $h^*(n) = dist(p, D)$ is the true remaining cost to reach delivery

$$\therefore \; h_2(n) \leq h^*(n)$$

so h_2 is admissible.

• Consistency Analysis

• Each landmark based term

$$h_l(p) = |dist(l, D) - dist(l, p)|$$

is consistent on unit cost graphs because shortest path distances are consistent and change by at most 1 between neighbours.

$$dist(l, p') \in \{dist(l, p) - 1, dist(l, p), dist(l, p+1)\}$$

so

$$h_l(p) \leq 1 + h_l(p')$$

Taking the max of consistent heuristics remain consistent ∴ h2 is consistent.

• Dominance comparison between h1 & h2

h_a dominates h_b if

$\forall n: h_a(n) \geq h_b(n)$ and $\exists n: h_a(n) > h_b(n)$ while both remain admissible

here • h2 uses obstacle aware shortest path info via landmark distances

• h1 ignores obstacles

⇒ In many grids h2 often strictly larger than Manhattan while still being a lower bound. For eg if a wall forces a long detour, Manhattan might be small but landmark lower bound captures path of detour length

So in practice; h2 dominates over h1 in such maps.

(#) How these Heuristics fit the Pareto-A* (L3) algorithm

In our search each node carries energy used so far (cost dimension) and reward collected so far (utility dimension)

h1 & h2 are used to lower bound the minimum remaining energy to reach delivery, enabling:

1. Feasibility Pruning:- if $h(n) > e$ then this node cannot reach delivery with remaining energy

2. Better Node ordering

use $g_{energy}(n) + h(n)$ as an optimistic completion cost for tie-breaking inside the Pareto frontier expansion policy

This gives us correctness and speed.